

GROUNDING PERSUASIVE LANGUAGE GENERATION FOR AUTOMATED MARKETING

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Paper under double-blind review

ABSTRACT

Effective use of persuasive yet truthful language is essential in marketing, from real estate listings to product advertisements. In this paper, we introduce a method to use LLMs for automated generation of such marketing content, using home listing descriptions as our focal application domain. Our agent is designed to align the generated content with user preferences while highlight useful factual attributes without generating misinformation. This agent is designed with three key modules: (1) Highlight Module, mimicking expert human behavior; (2) Personalization Module, aligning content with user preferences; (3) Grounding Module, ensuring factual accuracy and the inclusion of localized features. We validate our approach through comprehensive human-subject experiments in the domain of real estate marketing, demonstrating that the marketing descriptions generated by our approach surpass those original descriptions from human experts. Our findings provide a framework for leveraging LLMs to automate large-scale targeted marketing while addressing ethical considerations for responsible deployment of persuasive language generation technologies.

1 INTRODUCTION

While large language models (LLMs) have made significant strides across various tasks, their ability to persuade remains a critical but underexplored frontier. This is particularly striking given that persuasion — the common thread in almost all voluntary transactions from advertising and lobbying to litigation and negotiation — underpins roughly 30% of the US GDP (Antioch, 2013; McCloskey & Klammer, 1995). The potential to automate persuasion at scale offers unprecedented opportunities for boosting economic productivity across a wide range of sectors. Meanwhile, this same potential introduces serious safety concerns. If LLMs are able to generate persuasive content at scale, their influence on human opinions raises risks of misinformation, manipulation and misuse, especially in sensitive domains such as political campaigns (Voelkel et al., 2023; Goldstein et al., 2024).

In addition to these profound economic and societal implications, the relationship between the nature of intelligence and persuasion has been a fundamental research question since the time of early Greek philosophy. Aristotle viewed persuasion as both an art and an expression of intelligence, rooted in the ability to reason and communicate effectively. Yet the Greeks also cautioned against the sophistry that overly relies on rhetorical and emotional techniques divorced from the truth. This tension becomes especially relevant today with the rise of generative AI. Notable figures, including Sam Altman, have predicted that AI systems could achieve superhuman persuasion without superhuman general intelligence (Altman, 2023). This dichotomy raises several crucial questions for LLM research: How to reliably and consistently measure the fact-based persuasiveness? Does greater intelligence inherently lead to stronger persuasive capabilities? And if not, what specific abilities must LLMs develop to truly master persuasion? Surprisingly little is known about these questions, and this is what we embark on in this paper.

“The faculty of observing, in any given case, the available means of persuasion.”

— Aristotle, *Rhetoric*.

In this paper, we introduce *grounded persuasion* — a form of persuasion, inspired by Aristotle’s philosophy, that is grounded in fact, tailored to the audience, and adapted to contextual factors. Grounded persuasion is crucial for applications in marketing and advertising, and its effectiveness

can be linked directly to measurable behavior changes (e.g., in terms of engagement and conversions) while constrained by factual accuracy. We choose real estate marketing as our testbed for grounded persuasion and construct a realistic evaluation environment to capture the process of preference elicitation and measure the persuasiveness of preference-based generation. To operationalize grounded persuasion in this context, we design an LLM-based agent with three key modules: a Highlight Module, which mimics human expertise in identifying critical selling points; a Personalization Module, which tailors content to user preferences; and a Grounding Module, which ensures factual accuracy and integrates localized features. Through this model-based approach, we guide LLMs toward achieving superhuman persuasion in real estate marketing. Our findings lay the groundwork for leveraging LLMs in large-scale, targeted marketing and beyond, offering a scalable solution to complex persuasion tasks in real-world applications.

2 A BENCHMARK FOR GROUNDED PERSUASION

2.1 MOTIVATIONS AND CHALLENGES

The foundation of our study is built upon a consistent and effective evaluation benchmark. However, this effort faces several key challenges. One fundamental barrier is that persuasion is inherently subjective and heavily dependent on human feedback. Unlike many other LLM capabilities, such as reasoning, planning, or mathematical problem-solving, which have objective criteria for evaluation, persuasiveness lacks standardized metrics. The persuasiveness of a message is determined by its recipient and can vary greatly depending on individual preferences or the context in which the message is received. Therefore, any meaningful evaluation of LLM-generated persuasive content must incorporate human feedback to account for these complex and varied responses.

Another challenge comes from the multifaceted nature of persuasion — a process broadly studied in fields such as psychology, economics, and communication, each of which offers distinct models of how influence operates. Hence, the techniques required for effective persuasion can differ significantly across domains. In the realm of LLMs, most existing research has focused on political or opinion-based contexts, where persuasion often takes on an adversarial nature. These contexts complicate evaluation due to the strong influence of subjective beliefs and cognitive biases. For instance, studies like those conducted by Hackenburg & Margetts (2024) and Matz et al. (2024) reached differing conclusions regarding the effectiveness of LLMs in personalized persuasion, despite using similar experimental setups. Moreover, as noted by Durmus et al. (2024), opinion-based persuasion is susceptible to the anchoring effect, where participants remain strongly influenced by their initial beliefs, making it difficult to measure shifts in opinion accurately. Additionally, persuasion in these settings can often resort to deceptive strategies, as fact-checking is difficult even for experienced humans. Durmus et al. (2024) found that prompting models to fabricate information was the most effective strategy under current measures of persuasiveness. These challenges underscore the critical need to move beyond the scope of existing studies and develop new benchmarks that evaluate persuasion in more controlled, fact-based contexts.

Therefore, we develop an evaluation framework specifically designed for grounded persuasion. A crucial aspect of our design is to identify a domain that aligns well with grounded persuasion. We select real estate marketing as the primary testbed for several reasons. Firstly, the real estate sector is characterized by high-stakes economic decisions, where potential buyers tend to hold more fact-based, rational beliefs compared to political or emotionally charged domains. In this environment, persuasive language must not only resonate with potential buyers but also remain truthful and contextually relevant. This makes it an ideal setting for testing the principles of grounded persuasion. Secondly, real estate marketing involves complex decision-making processes, where strong persuasive capabilities can significantly influence outcomes. Experienced realtors are compensated through commissions that reflect the economic value of their persuasive skills. In addition, the potential of generative AI in this domain has been highlighted by a notable anecdote (User, 2023) that one claims a successful use case of LLM in facilitating a home sales without an agent. Thirdly, the availability of extensive datasets in the real estate sector enables us to extract valuable domain-specific knowledge for training LLMs and conducting thorough empirical evaluations. By leveraging these resources, our benchmark provides a robust and consistent means to assess LLMs’ grounded persuasion abilities, offering a scalable solution for evaluating persuasion in real-world, high-stakes scenarios.

2.2 A REALISTIC EVALUATION INTERFACE

We design the framework to ensure that the evaluation interface is realistic under two major criteria. First, we want the human subject to have an immersive experience with the context so that we gather authentic human feedback on the persuasiveness of the marketing content. Second, we need to elicit preferences of the human subject in a nature way such that we can properly test the dynamic generation of personalized content. Below we highlight a few key implementation details.

Content Curation To ensure that the content has both high quality and fidelity, we collect the real features of real estate listings on the market. The dataset for this experiment was sourced primarily from Zillow and includes around 50000 listings collected in the month of April in 2024. Each of these listings is from one of the top 30 most populous cities in the United States as described by the U.S. Census Bureau. Listings that were not residential in nature or were missing crucial data to this experiment were excluded from this dataset. This dataset is composed of 95 columns, with features ranging from number of bedrooms, price, views, and more (see Table 1 in Appendix D). These many features associated with each listing provide us sufficient space to develop and test improved models for grounded persuasion.

Preference Elicitation We design an interface to mimic the environment of online platforms that the model can observe the buyer’s general profile and behaviors (e.g., recently browsed or liked listing) to some degree. In our case of real estate listing, we ask the buyer to provide their preferences in a 1-5 scale on five general categories (price, location, home features & amenities, house size, investment value) and set a filter on the price range and number of bedrooms in the house they are looking for. This information allows us to select generally relevant listings to mitigate the anchoring effect that the marketing content can play little role to influence the buyer in the evaluation phase. In addition, we choose 5 relevant listings and ask the buyer to rate them on a 1-5 scale and provide their reasoning. This process ensures that we can collect a reasonable amount of each buyer’s preference information for the personalized persuasive content generation in the evaluation phase.

2.3 MEASURING THE FACT-BASED PERSUASIVENESS

To measure the performance in the task of grounded persuasion, we are particularly interested in how buyer behaviors are influenced by the generated content and whether it is factually accurate.

Evaluation Scheme Many previous work study persuasion by ask human how much does their opinion changes before and after reading an argument. In our task, human subjects often do not have any prior knowledge about item and this evaluation procedure would induce bias. Instead, we implement two alternative evaluation schemes: one is the A/B test where the buyer is presented with a single listing along with two descriptions generated by two distinct models and then asked to report which description makes them more interested in the listing; the other is the interleaved test where a set of listings each with a single description generated by some model and the buyer is asked to select the listings that they are interested in based on their descriptions.

Evaluation Metrics The persuasiveness is a relative concept so we employ the Elo score (Elo & Sloan, 1978), originally developed for ranking players in competitive games such as chess, to measure the relative persuasiveness of different text generated by humans and large language models (LLMs). The Elo rating system works by updating the scores of models after each comparison, adjusting based on the predicted outcome and the actual result. We also check whether the generated content is factually accurate. This is achieved by asking LLMs to extract the raw attributes from the description and compare them with the original data.

3 AI REALTOR

We introduce *AI Realtor*, an LLM-based agent that can use multi-level marketing information to compose persuasive descriptions for given real estate listings from the seller and can actively learn to fit personalized requirements for each buyer.

3.1 A MICRO-FOUNDATION OF OUR DESIGN

Our approach is built upon an microeconomic model of persuasion that is rooted from decision making and information economics, but is significantly adapted to be more practical for real-world marketing. Specifically, we represent a generic *product* X (e.g., a house on sale or a product on Amazon) as an n -dimensional vector $X = (X_1, X_2, \dots, X_n)$. Each X_i is called a *raw attribute*, which should be interpreted as raw and factual characteristics of the product (e.g., square footage of a house, floor number, distance to metro stations, etc.). With slight abuse of notation, a concrete product is denoted by vector $\mathbf{x} = (x_1, \dots, x_n)$ where $x_i \in \mathcal{X}_i$ is the value of raw attribute X_i . Let $\mathcal{X} = \prod_i \mathcal{X}_i$. During marketing, while sometimes attributes are mentioned, marketers more often describe *signaling features*, which often are attractive properties. In housing markets, signaling features include “large space”, “bright rooms”, “convenient locations”, etc. A crucial difference between raw attributes and signaling features is that attributes are factual hence not ambiguous, whereas signaling features are often ambiguous. To capture this difference, we let $S = (S_1, \dots, S_m)$ denote the set of all possible signaling features,¹ and $\mathbf{s} = (s_1, \dots, s_m)$ denote a realized vector value of these features. Specifically, we normalize $s_i \in [0, 1]$ to capture the *intensity* of signaling feature S_i for this product (alternatively, s_i can also be interpreted as probability of having S_i feature). For house listing applications, S_i could be “bright rooms” and correspondingly s_i denotes the extent to which this house has bright rooms. Notably, for a particular marketing domain, x_i ’s and s_j ’s can usually be identified by domain experts. For convenience, we often drop out the the word “raw” and “signaling”, and simply refer X_i as attributes and S_j as features when it is clear from the context.

Attribute-Feature Mapping. We assume there is an underlying mapping $\pi : \mathcal{X} \rightarrow [0, 1]^m$ that maps any product $\mathbf{x} \in \mathcal{X}$, represented by raw feature values $\mathbf{x} = (x_1, \dots, x_n)$, to its feature intensity values $\mathbf{s} = (s_1, \dots, s_m) \in [0, 1]^m$. That $\mathbf{s} = \pi(\mathbf{x})$. Sometime, we use $\mathbf{s}(\mathbf{x})$ to emphasize the dependence of $\mathbf{s}$ on the underlying attributes $\mathbf{x}$, and $s_j(\mathbf{x})$ is its j ’th entry. In the real-world, the mapping π is determined by humans’ common sense which tells, given factual values in $\mathbf{x}$ about the product, to an extent of s_j we can say the product has feature S_j . Therefore, we assume the mapping π is common knowledge, and known by every party in our model. One may wonder why marketers do not directly list all attribute values, but rather describe the signaling features. There are good reasons. From a practical point of view, the number of raw attributes is usually extremely large (e.g., there are hundreds of facts about a house), whereas each signaling feature may carry rich information about various features. For instance, a high value of feature “bright room” may implies multiple facts: a high floor, modern light system, large room size, sun-side facing, etc., though all probably. As can be seen, features tend to be vague — this is very often intentional due to the purpose of *signaling*. In particular, signaling is widely used in marketing, and more generally in persuasion, to shape agents’ beliefs using positive yet vague signals. Strategic signaling has been extensively studied in the literature of information economics, which some of our modeling has been drawing from.

Beliefs and Information Asymmetry. The intrinsic reason for marketing is due to the information asymmetry between sellers and buyers. That is, the seller or seller agent may perfectly know the product $\mathbf{x}$ and its feature values $\mathbf{s}(\mathbf{x})$ but a buyer enters the market with only a prior belief μ about the products in $\mathcal{X}$. Hence, lacking concrete detail of the particular product $\mathbf{x}$, the buyer begins with an expected belief of its features as follows:

$$\text{Initial feature belief of a buyer: } \bar{\mathbf{s}}(\mu) = \int_{\mathbf{x} \in \mathcal{X}} \mathbf{s}(\mathbf{x}) d\mu(\mathbf{x}) \quad (1)$$

Given the asymmetric feature beliefs between the buyer and seller, the purpose of marketing can be described as revealing features, subject to communication constraints, to shift buyer’s belief from $\bar{\mathbf{s}}(\mu)$ towards $\mathbf{s}(\mathbf{x})$ with the goal of increasing the product’s attractiveness to the buyer.

Grounded Persuasion based on Preferences and Surprises. While there have been standard economic models to capture buyers’ decision making and the persuasion process. These models turn out to be difficult to implement practically due to lack of concrete buyer utility functions and decision choices. Hence here we adopt a much more practical yet widely used approach in practice, i.e., the scoring method. At a high level, we will compute a score for each “grounded features”, i.e., those feature whose scores are high enough (formally, passing a pre-set threshold α), ensuring our

¹In real applications, this could be the set of typically mentioned features.

confidence of the product having this feature. Specifically, the set of grounded features is computed as followed, with a given threshold α

$$\text{Grounded Features: } S^\alpha(\mathbf{x}) = \{S_j : s_j(\mathbf{x}) \geq \alpha\} \quad (2)$$

Next, we score each $S_j \in S(\mathbf{x})$. Besides capturing how grounded the feature is, the score is designed to further captures two key tactics during persuasion — tailoring to buyer preferences and emphasizing surprising/uncommon features. Intuitively, a grounded feature S_j is *surprising* if it is among the top β -quantile of the distribution of S_j values under the prior distribution $s_j(\mu)$, or formally,

$$\text{Surprising Features: } SS^\beta(\mathbf{x}) = \{S_j \subset S : s_j(\mathbf{x}) \text{ is within } \beta\text{-quantile of distribution } s_j(\mu)\} \quad (3)$$

where β is a parameter. We further integrate the buyer preference by assuming a weight $r_j \in \mathbb{R}$ for each grounded feature S_j . In digital marketing applications, r_j can be learned through the standard industrial techniques of cookie analysis, though our implementation mostly relies on user surveys and GPT’s preferences as a proxy of humans’.

With the above, we simply define the score of each grounded feature $S_j \in S^\alpha(\mathbf{x})$ as $s_j(\mathbf{x}) + \lambda_1 r_j + \lambda_2 r_j \cdot \mathbb{I}(S_j \in SS^\beta(\mathbf{x}))$, where λ_1, λ_2 are weight parameters for preferences and surprises respectively. This additive form is due to our intuition of gradient-style updates which turns out to work very well in our real-world implementation, though we remark that this needs not to be the only choice in general so long as the score properly captures the dependences on preferences r_i and the value of surprises.

Consequently, our persuasion problem can be viewed as solving the following optimization problem to find the best k signaling features. These features will then be passed to GPT-4o as the prompt to generate a natural language description.

$$T^* = \arg \max_{T: |T| \leq k} \sum_{S_j \in S^\alpha} \left[s_j(\mathbf{x}) + \lambda_1 r_j + \lambda_2 r_j \cdot \mathbb{I}(S_j \in SS^\beta(\mathbf{x})) \right] \quad (4)$$

3.2 FROM MODELING TO ALGORITHMIC IMPLEMENTATIONS

From a high-level perspective, given a real estate listing $X = (x_1, x_2, \dots, x_n)$ containing n **raw features** $x_i \in \mathcal{X}$ (e.g., number of bedrooms), AI `Realtor` first employs a Highlight model (Section 3.3) to extract **highlights** $h(X) = (h_1, h_2, \dots, h_m), h_i \in \mathcal{H}$. Each highlight h_i (e.g., architectural styles in the listing) is a high-level abstraction of X , learned from past listing data to emphasize the most relevant features. This process serves as a pre-processing step to prepare the **sellers’** listings.

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Once the listing is prepared, the next phase focuses on the **buyers**. AI `Realtor` uses a local search engine to retrieve similar or relevant listings $N(Q) = \{\tilde{X}_1, \tilde{X}_2, \dots\}$ based on the buyer’s initial query $Q = (q_1, q_2, \dots, q_l), q_i \in \mathbf{X}$ (e.g., Price). This query specifies the buyer’s desired property filters. AI `Realtor` then constructs a user profile $U = \text{LLM}_{\text{gen}}([\mathcal{I}_{\text{Sum}}; V(q_1); V(q_2); \dots; V(q_l)])$ by summarizing the buyer’s stated preferences V into coherent natural language descriptions, where $V(q_i) \in \{1, 2, 3, 4, 5\}$ reflects the importance assigned to each query dimension q_i (e.g., $V(q_i) = 5$ signifies that “Price is the most important factor”) and $\mathcal{I}_{\text{Sum}}$ is the summarization prompt. These preferences are captured using a Preference Elicitation model (Section 3.4).

Finally, AI `Realtor` activates a Surprisal model (Section 3.5) to adjust the highlights $h(X)$ according to the user profile U , resulting in a personalized highlight plan $R(h(X), U)$. This personalized plan is then used to generate a final description $D_{\text{agent}} = \text{LLM}_{\text{gen}}([X; R(h(X), U)])$.

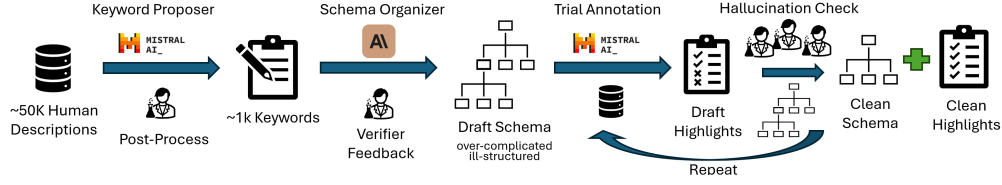


Figure 1: Illustration of the Highlight Schema Induction.

3.3 HIGHLIGHT MODULE: EXTRACTING MARKETING FEATURES

We model each item using its raw features X and high-level highlights $h(X)$. The raw features represent factual information available in the dataset, while each highlight is an abstraction derived from one or more raw features that buyers may consider during the decision-making process but are not directly observable. In fact, the complete set of highlights $h(X)$ is unknown. However, we observe that the human-written descriptions D_{human} on Zillow often implicitly convey user preferences for various highlights (e.g., whether it is convenient to commute). This offers a unique opportunity to construct a grounded and interpretable framework for automatic highlight extraction. We make the following assumption treating highlights as latent variables:

Assumption 1. $X \rightarrow h(X) \rightarrow D_{\text{human}}(X)$

Our task can thus be simplified as inferring $h(X)$ from observed $D_{\text{human}}(X)$.

To address this challenge, we first employ a human-AI collaboration to develop a **highlight schema** $\mathcal{T}$, which outlines the most common and useful highlight categories based on human-written descriptions in Zillow listings. Each category is annotated with keywords and examples to facilitate understanding.

Next, we split a portion of the dataset $\mathcal{D}_{\text{train}} \subset \mathcal{D}$ (detailed in Section 2.2) to serve as a training set. Using this training data, we guide a language model to extract highlights $h(X) = \text{LLM}_{\text{gen}}([\mathcal{T}; X])$, leveraging the schema $\mathcal{T}$ to ensure consistency and relevance in the highlights generated from each listing $X \in \mathcal{D}_{\text{train}}$ in the training set.

To mitigate potential hallucination issues during highlight extraction, we manually evaluate 636 highlight extraction results, refine the schema to improve clarity and precision, and re-run the model to re-generate the highlights. This process is repeated iteratively, with re-evaluations of the highlight outputs at each step. After several iterations, we find that the LLM’s extracted highlights generally align with user preferences. No significant hallucination issues were observed, thanks to the well-defined and thoroughly explained schema, though some aesthetic highlights extracted by the model may differ from human judgment. The complete pipeline is illustrated in Figure 1.

Highlight Schema Induction To induce the highlight schema $\mathcal{T}$, we first use *Mistral-8x7B-Instruct-v0.1* (Jiang et al., 2024) to extract keywords or phrases $\{k_1, k_2, \dots\} = \text{LLM}_{\text{gen}}([\mathcal{I}_{\text{Keyword}}; D_{\text{human}}])$ that summarize each human-written description D_{human} under a keyword-extraction prompt $\mathcal{I}_{\text{Keyword}}$ (Appendix B.1). We observed that, in some cases, the model output could not be directly parsed into a clean list of keywords, or it contained excessive quantifiers and modifiers. To address this, we re-prompted the model using $\mathcal{I}_{\text{Norm}}$ (Appendix B.2) to normalize each keyword. Through this process, we initially extracted 112,688 keywords—too many to handle effectively. We then applied additional normalization steps, including lowercasing, lemmatization, and synset merging via NLTK (Bird et al., 2009). We also filtered the keywords, retaining only those that appeared in at least 50 descriptions. This reduced the final set to 1,114 keywords.

Next, we prompted *Claude-3.5-Sonnet* (Anthropic, 2024) with a 100-keyword batch to iteratively generate a more complex hierarchical schema² that covers the majority of the keywords.³ We temporarily switched to *Claude-3.5-Sonnet* because we found it particularly difficult for open-source models, even the state-of-the-art *GPT-4o* (OpenAI, 2024a), to induce such a schema without group-

²The hierarchical structure helps participants better understand and organize highlight classes in an intuitive manner.

³An example run can be found in Appendix B.3.

ing most keywords into overly broad categories like "others" or "misc", resulting in a shallow and uninformative schema. In contrast, when fed keywords in small batches, *Claude-3.5-Sonnet* followed our instructions more faithfully, organizing the keywords into a carefully structured hierarchy. Every leaf node in the schema was associated with a set of relevant keywords. This process provided an initial schema draft, though it was still potentially over-complicated or poorly structured.

To evaluate the effectiveness of this schema for guiding LLM-based highlight extraction, monitor potential hallucination issues, and further refine the schema, we adopted a human-in-the-loop approach. We iteratively asked *Mixtral* to extract highlights based on the current schema $\mathcal{T}^{(t)}$ and human-written descriptions:

$$P(h(x)_i | D_{\text{human}}(X)) = \text{LLM}_{\text{gen}}(h(x)_i | [\mathcal{I}_{\text{ext_desc}}; \mathcal{T}^{(t)}; D_{\text{human}}(X)]) \quad (5)$$

Note here we make another assumption that **given human description, each highlight is conditionally independent**. The prompt $\mathcal{I}_{\text{ext_desc}}$ used here can be found in Appendix B.4.

Three human participants from our author team then reviewed hundreds of samples to determine whether hallucinations were present or if the schema needed further improvements.⁴ After two iterations and reviewing 636 samples, the participants helped refine the schema, making it more precise and adding more example keywords. They concluded that, under the latest iteration, no significant hallucination issues remained, although there were occasional disagreements on aesthetic highlights extracted by the model.

Highlight Extraction Dataset After inducing the highlight schema from the dataset, the next step is to build a Highlight Extraction model $P(h(X) | X; \theta)$, which directly predicts the distribution of highlights from raw features **without relying on human-written descriptions**. This is essential, as human-written descriptions are unavailable when a new listing is provided by a seller. To train this model, we adopt an imitation learning approach, grounded in the following assumption:

Assumption 2. *Listings with the highest ratio of $\#(\text{Likes})/\#(\text{Views})$ reflect population-level high-light preferences as expressed in their associated human-written descriptions.*

Specifically, we sort the listings based on the score $s(X) = \#(\text{Likes}(X))/\#(\text{Views}(X))$ in descending order and select the top 2,000 listings from our dataset as the highlight extraction dataset. This dataset is then randomly split into training and testing sets, following an 80:20 ratio. In this dataset, we assume human descriptions are available and thus use our description-conditioned feature extraction model $P(h(X)_i | D_{\text{human}}(X))$ to extract "Ground Truth" highlights. These "Ground Truth" highlights serve as labels for training our highlight extraction model $P(h(X) | X; \theta)$.

Highlight Prediction Model Since each listing X contains heterogeneous features (categorical, integer, float, etc.), we first convert each raw feature x_i into a natural language statement $nls(x_i)$ using the template, "The feature *feature_name* is *feature_value*."⁵ Next, we use an embedding model to convert each natural language statement into a fixed-dimensional vector $e_i = \text{LLM}_{\text{embed}}(nls(x_i)) \in \mathcal{R}^d$. We employ *SFR-Embedding-Mistral* (Meng et al., 2024) as the embedding model. The highlight model is then built as a simple multi-layer perceptron (MLP):

$$p_i = P(h(X)_i | X; \theta) = \sigma(O_i^T \text{ReLU}(W\bar{e}(X))) \quad (6)$$

where $\bar{e}(X)$ is the mean-pooled feature embedding, and $O_i \in \mathcal{R}^{d/2}$, $W \in \mathcal{R}^{d \times d/2}$ are the model's weights. The function σ represents the sigmoid activation function. Here, we assume conditional independence between highlights given the raw features X . Training is performed using logistic regression.

3.4 PERSONALIZATION MODULE: ALIGNING WITH USER PREFERENCE

This stage seeks to steer the text generation toward the buyer's preference. Specifically, after the highlighted attributes have been predicted at a population level, we adjusted the attribute scores according to each buyer's preference. However, as the marketing description in real world is not

⁴The annotation interface is shown in Appendix E.1.

⁵We exclude features such as "id", "likes", "views", "description", "url", "img_url", and missing features, as these are not relevant to highlight prediction.

optimized at a personalized level, we can no longer resort to the machine learning approach from the collected data. Instead, we leverage the innate capability of LLMs to understand and analyze human preference.

Structured Preference Representation As mentioned in Section 2.2, our environment allows us to elicit basic information from each buyer u . However, such information cannot directly describe the user’s preference. So, we ask the language model to act like a human realtor to determine the attributes that the users might be interested in based on their initial selection. To do this, we prompt the language model to convert the user preference into information structured according to the item attributes. We then ask the user to give a score $q_i(u)$ on a scale of 1-5 on how important each attribute $h(X)_i$ is along with their rationale.

Personalized Highlight Model With the user preference over each attributes, we adjust the attribute scores according to a heuristics below,

$$s_i = S(h(X)_i | X, u; \theta) = P(h(X)_i | X; \theta) + c(q_i - q_0), \quad (7)$$

where the constant c reflects the intensity of personal preference, q_0 is the basis rating of each attribute. In our experiment, we choose $c = 0.01$ and $q_0 = 2$. Finally, we choose 10 attributes with highest scores and list them in the prompt to generate persuasive marketing description (see a full specification in Appendix B.5).

3.5 GROUNDING MODULE: CAPTURING SURPRISAL VIA RAG

The last stage is designed to improve the text generation with more facts and valuable information signals. For each listing, we determine a set of attributes that have its comparative advantage among different groups of similar listings. In our case, we consider two kinds of retrieval criteria: (1) select all listings within the proximal location at different levels of granularity (e.g., neighbourhood, zipcode or city); (2) select the 20 listings with the most similar features via an information retrieval system (implemented by the ElasticSearch framework⁶) — the search engine implementation details can be found in Appendix C. For each group of similar listings, we determine an empirical distribution function on each attribute score $\tilde{F}_i$. We then set $1 - \tilde{F}_i(p_i)$ as the percentile ranking of the listing’s attribute i among this group. We then select all attributes that are among the top 30% percentile ranking for some group and provide the information in the prompt to generate persuasive marketing language (see a full specification in Appendix B.6). This gives the LLMs localized highlight information at different granularity level. We show an example from human feedback to support this finding in Appendix A.

4 EVALUATIONS

4.1 EVALUATION BY HUMAN FEEDBACK

Quality Assurance We use the Prolific platform to recruit participants for the human-subject experiments. We choose around 1000 listings in a variety of size and price range in the Chicago area and ask the participants to be in-state residents so that they have similar knowledge on the local housing market. To ensure the quality and consistency of human feedback, we implement several procedure in the human subject evaluation. First, we design a set of screening test to ensure that (1) participants are capable of understanding listing descriptions and extract information from them; (2) participants can follow a given home search motive to determine the more suitable home. Second, we include attention check, where over time we ask the human subject to compare two almost identical descriptions to make sure they are carefully reading the text and comparing their differences. Third, we implement a control experiment in which we ask each human subject to compare against the description generated by an LLM prompted to write plain descriptions that includes all the basic listing information. Lastly, we set up sufficient incentives in the evaluation platform to encourage more efforts spent on giving consistent human feedback. Specially, we ask for reasoning on their choices and make bonus payments to reward high quality responses.

⁶<https://www.elastic.co/elasticsearch>

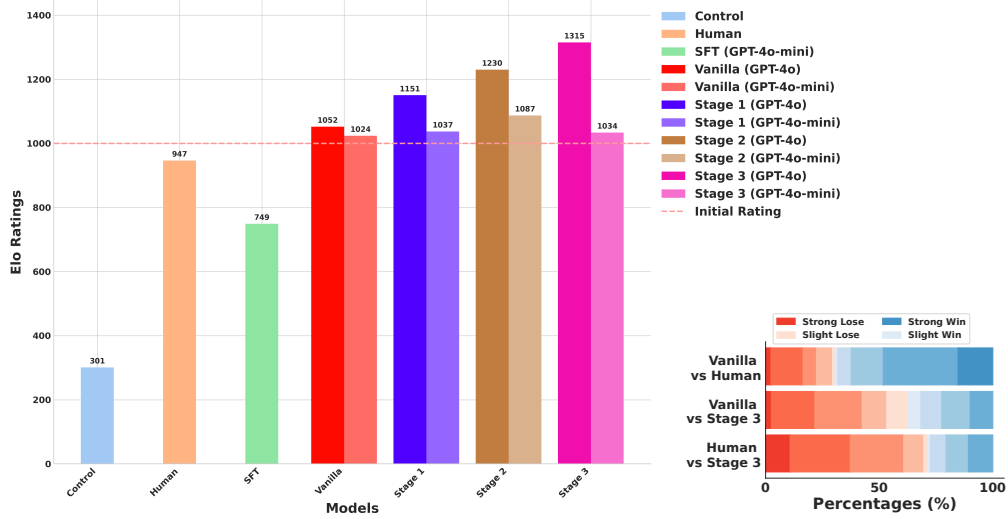


Figure 2: Model performance comparison using Elo ratings and empirical win rates

Performance Analysis We gather the human feedback on comparing the persuasiveness of listing descriptions generated by different models. These models includes: **Vanilla**, LLM prompted with all features of the listing; **Stage 1-3**, the proposed model from only the first stage to all three stages; **SFT**, LLM model after supervised fine-tuning and prompted with all features of the listing; **Human**, the listing descriptions on Zillow (written by professional realtors); **Control**, the model from the control experiment mentioned above. We also switch the underlying LLM models between GPT-4o and GPT-4o-mini, while keep the instructions in the prompt unchanged. Based on the pairwise comparison results, we compute the elo rating score of each model and illustrate them in Figure 2. We use typical choice of the initial Elo rating as 1000, scaling parameter $c = 400$ and learning rate $K = 32$, which corresponds to a model of win rate, $\frac{1}{1+10^{(r_1-r_0)/c}}$, for a model with elo rating r_0 against another with r_1 . We also plot the empirical win rate among three major models, where we use the intensity of color gradient to reflect the scale of relative persuasiveness given by the human subjects. From this result, we can observe a clear trend that each stage in our proposed model is taking some effect to improve the persuasiveness of the generated description. In addition, we also notice that only the more capable model, GPT-4o, is able to effectively leverage the highlight information provided in the prompt to generate more persuasive description.

4.2 EVALUATION THROUGH AI FEEDBACK

Human feedback can be costly, highly diverse, and difficult to reproduce. In this section, we investigate the potential of using AI feedback to simulate human survey participants.

Setup We assume that users adhere to their own preference profiles provided at the beginning of the survey (Section 4.1), and that the annotated samples independently represent these preferences. We use "GPT-4o-mini" (OpenAI, 2024b) as the simulator model and adopt the Sotopia framework (Zhou* et al., 2024) to configure this simulation agent. For each user, we provide the first four pairwise comparison samples as the few-shot training set and use the remaining six samples as the test set. User comments are added for the first four samples when available.⁷

Metrics Since this task involves binary classification for each pairwise comparison, we report Accuracy and F1 scores as the primary metrics. Additionally, we measure Spearman’s correla-

⁷An alternative setup directly uses the previous K comparison samples to predict the outcome of the $(K+1)$ comparison. However, we found the variance to be quite large (0.5). This highlights the challenge of on-the-fly evaluation, given the subjectivity and variability of human decision-making processes. We leave this issue of variance control for future work.

tion (Spearman, 1961) between the user’s actual preference ranking and the model’s predicted ranking, as recovering the true ranking may be more relevant in some cases.

Simulation Implementation The simplest approach is to directly prompt the model with the first four pairwise comparisons and ask it to predict the user’s choice for the subsequent comparison. However, our experiments show that this method yields poor performance (slightly better than 50%, which is close to random). We hypothesize that this occurs because the input contains too much structured information (mostly listing raw features), distracting the model from performing accurate reasoning.

To address this, we draw inspiration from the success of self-reflection prompting in decision-making tasks (Shinn et al., 2024). Specifically, we iteratively prompt the model to predict the user’s choice for the K -th ($K \leq 4$) pairwise comparison, review the output, and then ask the model to reflect on its own predictions. We augment the prompt with this reflection and predict the outcome of the $(K + 1)$ comparison without including the K -th listing. This process is similar to recursive summarization (Wu et al., 2021), where reflection on trial-and-error is used as a summary in each iteration.

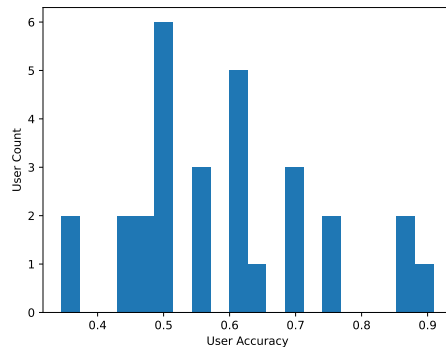


Figure 3: Simulation Accuracy histogram

Statistical Effectiveness of AI Feedback Our current best simulator model achieves an overall user-wide simulation accuracy of 65%, with the per-user simulation results illustrated in Figure 3. While the model demonstrates non-trivial ($\geq 60\%$) simulation performance for approximately 48.28% of users, there are still cases where the simulator performs worse than random in predicting user preferences, highlighting the inherent difficulty of user simulation.

Upon closer inspection, we find that certain users are particularly challenging to simulate. For instance, some users exhibit preferences for features that are either absent or difficult to infer from the listing data. This result underscores both the challenges and opportunities of utilizing AI feedback. On one hand, for users with relatively stable and well-defined preferences, our simulator offers a promising avenue for automatic benchmarking. On the other hand, for more demanding or indecisive users, accurately simulating their preferences is difficult, and real-time personalization remains a challenge, especially as their preferences may evolve over time.

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A SELECTED EXAMPLES

Below is an example where the user considers description B (generated by our proposed model) to be more persuasive, specifically because it includes localized features.

Description A: Welcome to your dream condo in the vibrant Printers Row neighborhood! This stunning one-bedroom, one-bathroom home is not only affordable but packed with modern amenities and stylish features, all for \$297,500. Nestled in the heart of Chicago, 547 S Clark St APT 405 offers unparalleled convenience with easy access to public transportation, shopping, dining, and the iconic Millennium Park. Built in 1998, this condo boasts a huge bedroom suite, hardwood flooring throughout, and an inviting gas fireplace. The newly upgraded stainless steel appliances and eye-catching granite countertops make the kitchen a chef's delight. Enjoy your morning coffee on the spacious private balcony, while additional storage and a walk-thru closet area cater to all your organizational needs. The property is equipped with central air and in-unit laundry for ultimate comfort and convenience. With a \$389 monthly HOA fee, this condo promises both affordability and a secure investment in the bustling heart of Chicago. Don't miss the opportunity to embrace a lifestyle of ease and style in this perfect one-bedroom sanctuary!

Description B: Discover the splendid charm of urban living at 547 S Clark St APT 405, a chic one-bedroom, one-bathroom condo in the heart of Chicago's desirable Printers Row. Priced at an enticing \$297,500, this condo harmonizes affordability with luxury, featuring an expansive 876 sqft layout adorned with elegant hardwood floors throughout and a cozy gas fireplace. Enjoy culinary delights with new stainless steel appliances and eye-catching granite countertops in the kitchen. **The huge bedroom suite boasts a walk-thru closet area, offering ample built-in cabinet space and additional storage—an rarity in similarly priced listings.** Revel in the tranquility of your spacious private balcony, perfect for unwinding with views of the bustling cityscape. Keep convenience at your fingertips with in-unit side-by-side laundry and parking in a secure garage. Built in 1998, this condo is a stone's throw away from Millennium Park and surrounded by public transportation, grocery stores, and cultural amenities, amplifying its urban attraction. With a manageable monthly HOA fee of \$389 and strong investment potential, this condo ensures financial peace of mind alongside lifestyle enrichment. Don't miss out on this prime opportunity in vibrant Cook County, where your ideal location meets impeccable structure and style.

User Feedback: I selected Description B because while both descriptions effectively showcase the property's appealing features, Description B employs a more evocative and sensory language, creating a stronger emotional connection and a more vivid image of the lifestyle the condo offers. Specific points of comparison:

- Language & Tone:

Description B uses words like "splendid charm," "chic," and "harmonizes" to paint a picture of elegance and sophistication, creating a more aspirational tone.

Description A, while positive, uses more straightforward language, focusing on practicality and convenience.

- Emphasis on Space & Luxury:

Description B highlights the "expansive 876 sqft layout," and the "huge bedroom suite," emphasizing the sense of space and luxury.

Description A also mentions the spaciousness but doesn't create as strong an image of grandeur

- Unique Selling Points:

Description B specifically points out the rarity of the ample storage and built-in cabinetry in similarly priced listings, making the property stand out.

Description A focuses on the general convenience and modern amenities, which, while attractive, are not as unique.

- Lifestyle & Surroundings:

Description B paints a more vivid picture of the lifestyle the condo offers, inviting the buyer to "revel in the tranquility" of the balcony and highlighting the proximity to cultural amenities, creating a stronger sense of place.

Description A mentions the location and amenities but lacks the same level of detail and emotional connection.

- Overall:

Both descriptions are well-written and informative, but **Description B's richer language, focus on unique features, and emphasis on lifestyle create a more compelling and emotionally resonant picture of the property.** It makes the condo feel more desirable and aspirational, which is likely to attract more potential buyers.

B PROMPTS

B.1 KEYWORD EXTRACTION PROMPT

```

Your task is to extract attractive highlights. (e.g., 'modern
  → amenities', 'great views', 'lush landscaping', 'bamboo
  → flooring'). Please express these highlights as phrases or
  → keywords from the following house description. Each
  → highlight should be separated by a comma. \n\nDescription: {
  → desc}\n\nHighlights:

```

B.2 KEYWORD EXTRACTION NORMALIZATION PROMPT

```

Please remove the quantifiers, numbers, adjectives or any
  → modifiers in the provided phrase. "
Uppercase or lowercase doesn't matter. "
If the given input is already precise enough, please provide
  → the same input.
If you are not sure what to do, please also provide the input
  → as it is. "
Do not explain or provide additional information.
Here are a few examples:
\n\nInput: Two Bedrooms.\n\nOutput: Bedrooms.
\n\nInput: Newly Renovated Kitchen.\n\nOutput: Kitchen.
\n\nInput: landscape. \n\nOutput: landscape.
[Example Ends]
Now, given the Input, please precisely provide the Output.
\n\nInput: {} \n\nOutput (should only be a noun phrase or
  → keyword): "

```

B.3 SCHEMA INDUCTION PROMPT

```

Here is an initial listing keyword schema that I have, but it may
  → not be comprehensive. I have a manually extracted
  → comprehensive keyword list, but there are many duplicated
  → words (e.g., different keywords may bear similar semantic
  → meanings) and some of them may inspire new categories in
  → this schema. I will give you that 1k+ keyword list in a file
  → and the schema below. Can you do it this way: for every 100
  → keywords in the file, either try to assign it in one of the
  → categories below, or create a new (sub)category and assign
  → the keyword to this new (sub)category. You CANNOT use too
  → broad categories like "others" "misc" "uncategorized". Only
  → create informative categories if necessary. Give me the
  → final zip files containing all 100-ish intermediate
  → assignment results. Each result should be represented as a
  → JSON-like file with key=subcategory, value=[
  → list_of_original_keywords_in_file], or key=category, value=
  → subcategory (in other words, I want a rich hierarchical
  → structure with the leaf nodes as list of original keywords
  → in file).

```

```

###schema###
Appliances:
  Refrigerator
  Oven
  Dishwasher
  Washer/Dryer

```

702 Microwave
 703 Garbage Disposal
 704
 705 Transportation:
 706 Garage
 707 Carport
 708 Parking Space
 709 Public Transit Nearby
 710
 711 Interior Features:
 712 Hardwood Floors
 713 Fireplace
 714 Central Air Conditioning
 715 Walk-in Closet
 716 Open Floor Plan
 717 High Ceilings
 718
 718 Exterior Features:
 719 Balcony
 720 Patio
 721 Deck
 722 Fenced Yard
 723 Garden
 724 Pool
 725
 725 Building Features:
 726 Elevator
 727 Fitness Center
 728 Laundry Room
 729 Security System
 730 Concierge
 731
 731 Utilities:
 732 Water
 733 Gas
 734 Electricity
 735 Cable/Satellite TV
 736 Internet
 737
 738 Neighborhood Features:
 739 Nearby Schools
 740 Parks
 741 Shopping Centers
 742 Restaurants
 743 Hospitals
 744 Recreation Facilities

746 B.4 FEATURE EXTRACTION BASED ON DESCRIPTION PROMPT

747 "Your task is to determine whether the given label class category
 748 ↪ is mentioned in the description. The meaning of the category
 749 ↪ will be explained by example keywords in this category.
 750 ↪ Only respond by 'YES' or 'NO'. "
 751 "Label Class Category: {highlight_class}. \n\nExample Keywords in
 752 ↪ this Category: {keywords}\n\n"
 753 "\n\nDescription: {human_description}\n\nResponse (Yes/No): "
 754

755 B.5 PERSUASIVE LANGUAGE GENERATION WITH PERSONALIZED HIGHLIGHTS

"Your task is to generate a marketing description for a real
 ↳ estate listing with the provided features, a list of
 ↳ attributes to highlight, and the client's preferences.
 - The listing has the following features:\n{
 ↳ features_description}
 - The listing has the following attributes (accounted for the
 ↳ client's preference) that are worth highlighting:\n{
 ↳ highlight_features_reweigh }
 - The client has the following general preferences:\n{
 ↳ user_preference}
 - The client has the following specific preferences over
 ↳ attributes:\n
 {attribute_preference}
 - You should emphasize the feature or attributes that matches
 ↳ with the user's preference.
 Make sure the description is persuasive while concise under
 ↳ one paragraph."

B.6 PERSUASIVE LANGUAGE GENERATION WITH LOCALIZED FEATURE PROMPT

"Your task is to generate a marketing description for a real
 ↳ estate listing with the provided features, a list of
 ↳ attributes to highlight and a list of attributes that are
 ↳ competitive among similar listings."
 - The listing has the following features:\n{
 ↳ features_description}
 - Compared with {K} similar listings, the listing stands out
 ↳ in the following attributes that you want to emphasize:
 {surprisal_attributes}
 - Compared with listings in Chicago, the following attributes
 ↳ of this listing are competitive:\n
 {city_rankings}
 - Compared with listings in this neighborhood {neighbourhood},
 ↳ the following attributes of this listing are
 ↳ competitive:\n
 {neighbourhood_rankings}
 - Compared with listings in this zipcode {zipcode}, the
 ↳ following attributes of this listing are competitive:\n
 {zipcode_rankings}
 - Finally, You should explicitly highlight the listing
 ↳ features or attributes that stands out above or those
 ↳ ones that exactly matches with the user's preferences as
 ↳ a surprise factor.
 Make sure the description is persuasive while concise under
 ↳ one paragraph."

C RETRIEVER IMPLEMENTATION DETAILS

In this section, we will explain the implementation details of our retrieval system.

C.1 SEARCH ENGINE SCHEMA

```
"mappings": {
  "properties": {
    "bedrooms": {"type": "float"},
    "bathrooms": {"type": "float"},
    "price": {"type": "float"},
```

```

810         "description": {"type": "text"},
811         "area": {"type": "float"},
812         "street_address": {"type": "text"},
813         "home_type": {"type": "keyword"},
814         "state": {"type": "keyword"},
815         "city": {"type": "keyword"},
816         "page_view_count": {"type": "float"},
817         "favorite_count": {"type": "float"},
818         "home_insights": {"type": "keyword"},
819         "neighborhood_region": {"type": "keyword"},
820         "id": {"type": "keyword"}
821     }
822 }

```

D DATA CURATION

D.1 DATASET SCHEMA

Table 1: Listing data, subset of important columns

Field Name	Data Type
bedrooms	float64
bathrooms	float64
price	float64
description	object
living_area_value	float64
lot_area_value	float64
area_units	object
brokerage_name	object
zipcode	object
street_address	object
home_type	object
time_on_zillow	object
page_view_count	float64
favorite_count	float64
home_insights	object
neighborhood_region	object
scraped_at	object
url	object
city	object
state	object
year_built	float64
county	object
avg_school_rating	float64
id	object
time_on_zillow_days	float64
score	float64
jpeg_urls	array

E INTERFACE

E.1 FEATURE HIGHLIGHT ANNOTATION INTERFACE

We design an interface for the annotator to select the features highlighted in a given description and to provide feedback on the quality of extractions by LLMs. See Figure 4.

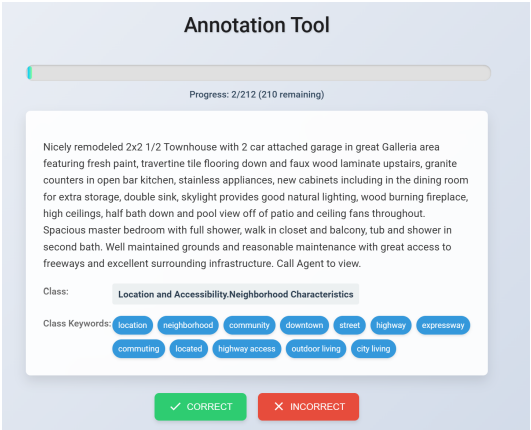


Figure 4: Feature Highlight Annotation Interface